

CSCI 6360

Project 3

April 27, 2026

1 Random Forest Performance on Dataset 2

The Random Forest model was trained and evaluated on both in-sample and out-of-sample data. Hyperparameter tuning was performed using cross-validation on the training set to determine the optimal parameters. The tuning process considered the following parameters:

```
params = {  
    'n_estimators': [200, 300, 400, 500, 700],  
    'max_depth': [10, 20, 30, None],  
    'min_samples_split': [2, 5, 10],  
    'min_samples_leaf': [1, 2, 4],  
    'max_features': ['sqrt', 'log2', 0.5],  
    'bootstrap': [True, False]  
}
```

2 In-Sample Quality of Fit Metrics

Table 1: In-sample performance metrics of the Random Forest model.

Metric	Value
R^2	0.9549
SST	9.78×10^{11}
SSE	4.41×10^{10}
MSE	441081.88
RMSE	664.14
MAE	374.50
m	100000

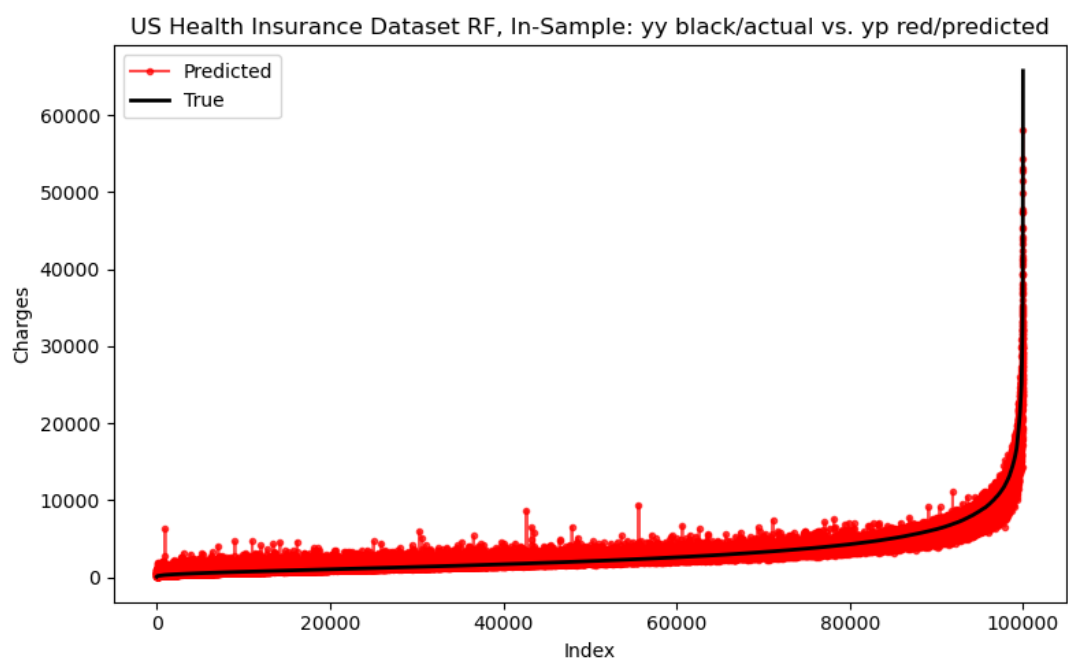


Figure 1: In-sample performance of the Random Forest model on Dataset 2.

3 Out-of-Sample Quality of Fit Metrics

Table 2: Out-of-sample performance metrics of the Random Forest model.

Metric	Value
R^2	0.6510
SST	1.97×10^{11}
SSE	6.87×10^{10}
MSE	3434098.68
RMSE	1853.13
MAE	1025.39
m	20000

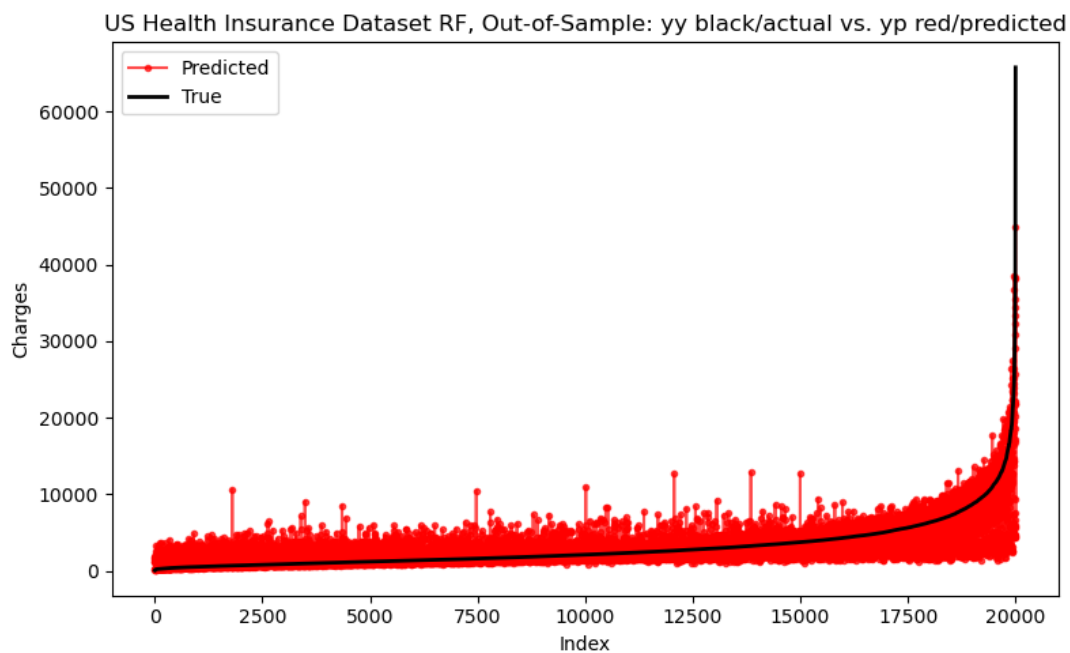


Figure 2: Out-of-sample performance of the Random Forest model on Dataset 2.

4 Fine-Tuned Quality of Fit Metrics

A total of 20 hyperparameter configurations were randomly sampled, and each was evaluated using 5-fold cross-validation. The following parameter settings were identified as optimal:

```
Best params: {'n_estimators': 300,
'min_samples_split': 5, 'min_samples_leaf': 4,
'max_features': 0.5, 'max_depth': None,
'bootstrap': False}
```

Table 3: Fine-tuned performance metrics of the Random Forest model.

Metric	Value
R^2	0.6542
SST	1.97×10^{11}
SSE	6.80×10^{10}
MSE	3402240.89
RMSE	1844.52
MAE	999.62
m	20000

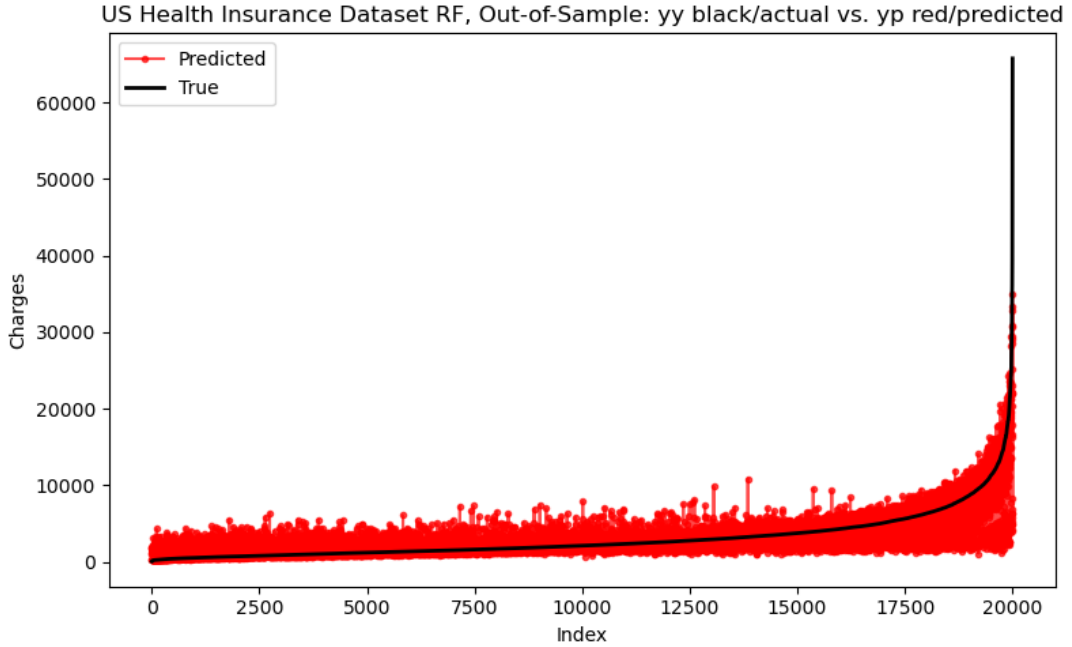


Figure 3: Fine-tuned performance of the Random Forest model on Dataset 2.

5 Feature Importance from the Best-Performing Model

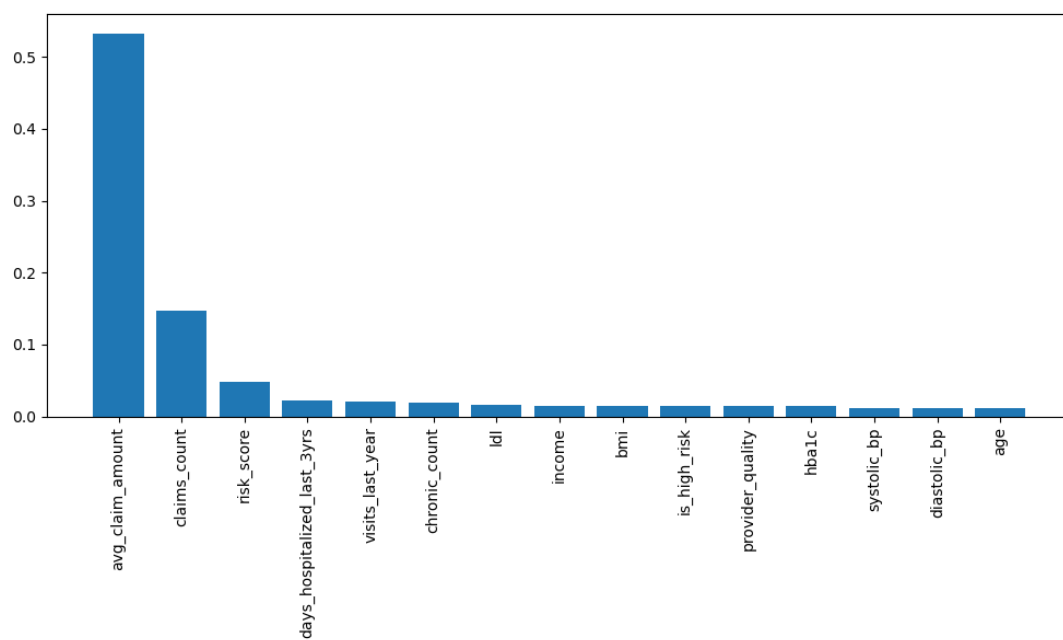


Figure 4: Top 15 Most Important Features.